Auto-parts

A bank owns a strip-mall, located along a busy street (say Sagamore Parkway in West Lafayette). It rents space in the mall to various stores. Not all locations in a strip-mall are equally attractive since not all are equally visible from the street. The crucial distinction is whether or not the location is an **end-cap**. If one thinks of a strip mall as U-shaped, the tips of the U abut the street. The end-cap locations are exactly those two tips of the U.

In early 2011, an auto-parts chain signed a lease with the bank to occupy one of the two end-cap locations in the strip mall. In late 2011, soon after the auto-parts store moved in, the bank started construction of a building, which partially obstructed the view from the street to the auto-parts store. After a bit more than a year of operating the store and some negotiations between the auto-parts company and the bank, the former sued the latter for breaking its contract, claiming that the location was not an end-cap location. The auto-parts company defined an end-cap location as one that is easily visible from the street.

To demonstrate that the location was not an end-cap one, the auto-parts company claimed that its 2012 sales in this location were much lower than they would have been had it been an end-cap location. The following information is provided:

- The 2012 sales at this store is $1,883,000.
- The population within a 3-mile radius of this location is 66,000.
- The average annual income in a 3-mile radius of this location is $70,000.
- The average Yelp rating of the store is 2 stars.
- Data on 45 other auto-parts stores in other strip-mall location in the state of Indiana (see the file *autoparts.xls*). For each store in the file, you know whether or not the location is an end-cap (1 means end-cap, 0 means not), sales in 2012, population size in a 3-mile radius as well as the average annual income in a 3-mile radius and its Yelp-rating score.

Part 1 (Descriptive Statistics)
You are hired by the auto-parts company (or by the bank) to investigate this claim to possibly support (or refute) it. Pick a side and use descriptive statistics to argue with your classmates about the claim.

- What would you calculate from this data to support or refute the argument that sales at this store is below average among the end-cap stores?
- What other information in this data can potentially support the argument that sales at this store is low?

Part 2 (Inferential Statistics, to be discussed in later class)

Use regression models and dummy variables to solve the case.

- Run a regression of sales on all other variables.
- Is there significant linear relationship among all the variables? Which one is most significant?
- Test the joint significance of population and income.
- Run residual analysis to validate the model.
- Use interaction terms to investigate the "impact" of yelp rating between end-cap and non-cap locations. Is there a significant difference about this impact between the two groups?
- Select a reasonable model and provide the 95% confidence and prediction interval for the sales of a store with similar parameters.
- Conclude the case.
- How would your conclusion change if the store has 2.5 or 3 stars on Yelp?